

A Machine-Checked Reduction of the Riemann Hypothesis to Pólya–Frequency Positivity of the ξ Taylor Coefficients

Biswajit Mondal

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Abstract

We present a formally verified study, in Lean 4 / Mathlib, of the Laguerre–Pólya / total-positivity approach to the Riemann Hypothesis (RH). **We do not prove RH.** We prove a chain of *conditional reductions*, each checked by the Lean kernel and depending only on the foundational axioms `{propext, Classical.choice, Quot.sound}`, that convert RH into total-positivity statements about an explicit positive kernel and into an operator-positivity (pseudo-Hermitian) statement. Our central result is a *tightness theorem*: modulo three classical, machine-uncertified but standard inputs (Pólya–Jensen, Aissen–Schoenberg–Whitney/Edrei, and the Laguerre–Pólya closure), RH is *equivalent* to total positivity of the Toeplitz matrix of the sign-normalized ξ Taylor coefficients. Consequently this lone input cannot be “postulated” as progress: doing so is logically identical to assuming RH. Along the way we identify and repair a target-function error (building the program on the entire completed zeta Λ_0 , whose zeros are *not* the Riemann zeros, rather than on Riemann’s ξ). We complement the formal results with an unusually extensive high-precision computation: the ξ Taylor coefficients $b_0, \dots, b_{120}$ (with $|b_{120}| \approx 10^{-449}$), cross-certified to over 230 significant digits, verify the reduction’s premise at scale—all contiguous Toeplitz (Pólya-frequency) minors up to order 16 are strictly positive across all offsets, the order-2 sequence is log-concave for all $n \leq 118$, and all Jensen polynomials of degree ≤ 6 are hyperbolic up to $n = 114$. This range substantially exceeds the order-2 (Turán) and modest-degree Jensen checks common in the literature, and localizes the open content to *uniformity*. The development comprises 47 Lean modules verified by a discipline that rejects `sorry` and added axioms.

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1 Introduction and scope

Disclaimer. This paper is a *formalization and reduction study*, not a proof of the Riemann Hypothesis, and it makes no claim to be one. Every implication labelled “verified” is checked mechanically; every classical theorem we rely upon but do not formalize is cited explicitly as an *external input* and is never silently assumed. The defensible contributions are: (i) a correct, machine-checked, and *tight* reduction of RH to a single sign-correct positivity statement; (ii) the identification and repair of a target-function error; (iii) an independent operator-positivity reduction; and (iv) reproducible numerical evidence delimiting the open frontier.

The object. Riemann’s entire ξ function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) \quad (1)$$

is entire, and its zeros are exactly the nontrivial zeros of ζ ; RH asserts they all lie on $\operatorname{Re} s = \frac{1}{2}$. On the critical line, the real even function $\Xi(t) = \xi(\frac{1}{2} + it)$ has a Taylor expansion

$$\Xi(t) = \sum_{n \geq 0} b_n t^{2n}, \quad b_n \in \mathbb{R}, \quad (2)$$

and we set the *sign-normalized sequence* $\mu_n := (-1)^n b_n$.

Two classical pillars.

- *Pólya–Jensen* [3, 7]: RH is equivalent to hyperbolicity (all roots real) of every Jensen polynomial $J_{d,n}(X) = \sum_{k=0}^d \binom{d}{k} b_{n+k} X^k$.
- *Aissen–Schoenberg–Whitney / Edrei* [4, 5, 6]: a real sequence is a *Pólya-frequency* (PF) sequence—all minors of the Toeplitz matrix $[a_{i-j}]$ are nonnegative—iff its generating function is $e^{\gamma z} \prod (1 + \alpha_i z) / \prod (1 - \beta_j z)$ with $\gamma, \alpha_i, \beta_j \geq 0$; for an entire generating function this is exactly Laguerre–Pólya membership.

Together: $RH \iff (\mu_n) \text{ is a Pólya-frequency sequence.}$

Pólya’s positive kernel. By Pólya’s representation [2], $\Xi(t) = 2 \int_0^\infty \Phi(u) \cos(ut) du$ with $\Phi > 0$, so

$$b_n = (-1)^n \frac{2M_n}{(2n)!}, \quad M_n := \int_0^\infty \Phi(u) u^{2n} du > 0, \quad \mu_n = \frac{2M_n}{(2n)!} > 0. \quad (3)$$

2 The target-function correction: ξ , not Λ_0

We record an error and its repair, because catching it is part of the contribution. Mathlib’s *entire* completed zeta $\Lambda_0 = \text{completedRiemannZeta}$ satisfies $\Lambda = \text{completedRiemannZeta} = \Lambda_0 - 1/s - 1/(1-s)$.

Proposition 1 (Wrong target). *The zeros of Λ_0 are not the nontrivial zeros of ζ . Indeed, at a nontrivial zero ρ one has $\Lambda(\rho) = 0$ but $\Lambda_0(\rho) = \frac{1}{\rho} + \frac{1}{1-\rho} \neq 0$. On the critical line, $\Lambda_0(\frac{1}{2} + it) = (1 - 2\Xi(t))/(t^2 + \frac{1}{4})$, whose zeros form the level set $\{\Xi = \frac{1}{2}\}$, unrelated to RH.*

The remedy is to remove the poles of Λ by *multiplying* by $s(s-1)/2$ (which preserves zeros), recovering ξ of (1). We verify the corrected slice has the correct zero set:

Theorem 2 (Xi_eq_zero_iff_riemannZeta; verified). *For all $t \in \mathbb{R}$, $\Xi(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$.*

Proof (mechanized). We define $\Xi(t) = -\frac{t^2+1/4}{2} Z(t)$ where $Z(t) = \text{Re } \Lambda(\frac{1}{2} + it)$ is the verified real critical-line slice. The scalar $-\frac{t^2+1/4}{2}$ is a nonzero real, so $\Xi(t) = 0 \iff Z(t) = 0 \iff \Lambda(\frac{1}{2} + it) = 0 \iff \zeta(\frac{1}{2} + it) = 0$ (the last step using $\Lambda = \Gamma_{\mathbb{R}} \cdot \zeta$ with $\Gamma_{\mathbb{R}}(\frac{1}{2} + it) \neq 0$). All steps are checked in Lean. $\square$

All total-positivity, Jensen, and moment results below are stated for this corrected ξ .

3 Verified reductions

Throughout, “verified” means the Lean theorem reduces to $\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$, confirmed by `##print axioms`. Definitions of type `Prop` that encode classical theorems we do not formalize are *named external inputs*; they are never asserted, only taken as explicit hypotheses.

3.1 The Pólya-frequency / Toeplitz route

Definition 3 (XiToeplitzTotalPositive). The sign-normalized sequence (μ_n) is *Pólya-frequency total positive* if every minor of the Toeplitz matrix $[\mu_{i-j}]$ (over all strictly monotone row and column index selections) is nonnegative.

Proposition 4 (Low-order rungs; verified). *Under XiToeplitzTotalPositive: (i) every 1×1 minor gives $\mu_n \geq 0$; (ii) the contiguous 2×2 minor equals the Turán determinant, $\mu_{n+1}^2 - \mu_n \mu_{n+2}$, so its nonnegativity is exactly the Turán inequality $b_n b_{n+2} \leq b_{n+1}^2$; (iii) for every order k and offset m the contiguous minor $\det[\mu_{m+i-j}]_{i,j < k} \geq 0$.*

We further prove (Lemma `xiToeplitzMinor3_sign_eq`) that the order-3 signed minor equals $(-1)^n$ times the unsigned b -determinant, exhibiting the correct alternating-positivity structure; and we document that the Hankel/moment route used in an earlier draft is the *wrong* machine, since Hankel positivity forces *log-convexity*, the opposite of the (true) Turán log-concavity.

3.2 Tightness of the reduction

Let the three classical external inputs be: `XiToeplitzTPToScaledFiniteBridge` (Edrei–ASW), `XiLaguerrePolyaClosureBridge` (LP closure), and `PolyaJensenBridge`; and let `XiToeplitzTPFromRH` be the classical converse ($\text{RH} \Rightarrow \xi$ is $\text{LP} \Rightarrow$ its coefficients are PF).

Theorem 5 (`xiToeplitzTotalPositive_iff_riemannHypothesis`; verified). *Given the three classical bridges and the classical converse,*

$$\text{XiToeplitzTotalPositive} \iff \text{RH}.$$

Remark 6 (The honesty linchpin). Theorem 5 shows the lone non-classical input is *equivalent* to RH. Hence declaring it an axiom is logically identical to `axiom rh : RiemannHypothesis`: it does not constitute a proof, and `##print axioms` would expose the assumption. A reduction is progress only if its hypothesis is *weaker than, or independent of*, RH; this one is provably neither.

3.3 The complete moment-determinant ladder

Under Pólya's representation (the named external input `XiMomentKernelRep`, encoding (3)), the PF condition transfers to explicit moment determinants at *every* order.

Theorem 7 (`xiToeplitzContigMinor_eq_of_kernelRep`; verified). *For all $k, m \in \mathbb{N}$,*

$$\det[\mu_{m+i-j}]_{i,j < k} = 2^k \cdot \det\left[\frac{M_{m+i-j}}{(2(m+i-j))!}\right]_{i,j < k}.$$

Consequently (`xiToeplitzContigMinor_nonneg_iff_moment_of_kernelRep`) Pólya-frequency positivity at all orders is equivalent to positivity of the factorial-weighted moment determinants of the explicit positive kernel Φ .

Proof (mechanized). Each entry of the signed Toeplitz matrix equals 2 times the corresponding factorial-weighted moment entry (by (3)); the determinant is then 2^k times the moment determinant by multilinearity (`Matrix.det_smul`, `#Fin k = k`). Positivity of 2^k gives the equivalence. $\square$

The order-2 mechanism is quantified by the *slack factor* $s_n = \frac{(2n+3)(2n+4)}{(2n+1)(2n+2)}$, for which we verify $s_n > 1$ (`one_lt_slackFactor`) and $s_{n+1} < s_n$ (`slackFactor_succ_lt`); thus $s_n \downarrow 1$.

3.4 Order-2 unification: Turán $\Rightarrow$ Jensen hyperbolicity

Theorem 8 (`jensenPoly_two_hyperbolic_of_turan`; verified). *If $b_{n+2} \neq 0$ and the Turán inequality $b_n b_{n+2} \leq b_{n+1}^2$ holds, then the degree-2 Jensen polynomial $J_{2,n}$ is hyperbolic (all complex roots real). Hence (`allJensenTwo_hyperbolic_of_turan2All`) if all coefficients are nonzero and all Turán inequalities hold, every $J_{2,n}$ is hyperbolic.*

Proof (mechanized). $J_{2,n}$ maps to $b_{n+2}z^2 + 2b_{n+1}z + b_n$ over $\mathbb{C}$. Completing the square, $(b_{n+2}z + b_{n+1})^2 = b_{n+1}^2 - b_n b_{n+2} =: D \geq 0$ at any root. The auxiliary lemma `im_zero_of_sq_eq_nonneg_real` ($w^2 = D \in \mathbb{R}_{\geq 0} \Rightarrow \text{Im } w = 0$) forces $b_{n+2}z + b_{n+1} \in \mathbb{R}$, hence (as $b_{n+2} \neq 0$ real) $\text{Im } z = 0$. No square roots and no zero-locus input are used. $\square$

Nondegeneracy $b_{n+2} \neq 0$ is supplied by (3) ($M_n > 0$).

3.5 An independent operator-positivity route

The naive Hilbert–Pólya operator must be self-adjoint for the ambient inner product—the obstruction. The pseudo-Hermitian viewpoint [10, 11] keeps only a *positive metric*.

Definition 9 (`PseudoHermitianForm`). For a $\mathbb{C}$ -linear T on a $\mathbb{C}$ -vector space E , a pseudo-Hermitian form is a map $\eta: E \times E \rightarrow \mathbb{C}$, linear in the first slot and conjugate-linear in the second, with $\eta(Tx, y) = \eta(x, Ty)$ (η -self-adjointness) and $\eta(x, x) \neq 0$ for $x \neq 0$ (nondegeneracy, supplied by positive-definiteness).

Theorem 10 (`PseudoHermitianForm.eigenvalue_conj_eq` and `riemannHypothesis_of_pseudoHermitianWithn` verified). *An η -self-adjoint operator has real eigenvalues. Consequently, if the critical-strip zeros are realized as the η -self-adjoint spectrum via the encoding $\mu(s) = -i(s - \frac{1}{2})$ (real iff $\text{Re } s = \frac{1}{2}$), then RH holds.*

Proof (mechanized). For an eigenpair $Tv = \mu v$, $v \neq 0$: $\mu \eta(v, v) = \eta(Tv, v) = \eta(v, Tv) = \bar{\mu} \eta(v, v)$, and $\eta(v, v) \neq 0$, so $\mu = \bar{\mu}$. The proof uses only the form laws—no inner product, completeness, or boundedness—so this route is genuinely independent of the symmetric one. Existence of (E, T, η) with positive η and the prescribed spectrum is open. $\square$

4 Numerical evidence

We computed the ξ Taylor coefficients $b_0, \dots, b_{120}$ (hence μ_n) by a numerically stable Cauchy-circle extraction and *cross-certified* them at two working precisions (620 and 520 digits, distinct contour radii): all coefficients agree to at least 236 significant digits over the whole range, with $|b_{120}| \approx 10^{-449}$. Analysis (minors, roots) is carried out at 120 digits. Code: `scripts/rh_report_figures.py` (mpmath). Determinants use arbitrary-exponent arithmetic, so the extreme dynamic range (minors reach $\sim 10^{-6000}$ at order 16) is exact in scale, not underflowed.

Scale of the verification. Within the certified range we find: (i) $\mu_n > 0$ for all $n \leq 120$; (ii) the order-2 Turán inequality, indeed strict log-concavity $\mu_{n+1}^2 > \mu_n \mu_{n+2}$, for all $n \leq 118$; (iii) *every* contiguous Toeplitz (Pólya-frequency) minor of order $k \leq 16$, at all offsets, is strictly positive; (iv) every Jensen polynomial of degree $d \leq 6$ is hyperbolic for all $n \leq 114$ (worst observed $|\text{Im root}| \sim 10^{-320}$, i.e. numerical zero). To our knowledge this is the most extensive direct verification of the full Toeplitz/PF total-positivity ladder for ξ reported to date; prior numerical work centres on the order-2 (Turán) inequalities and moderate-degree Jensen hyperbolicity.

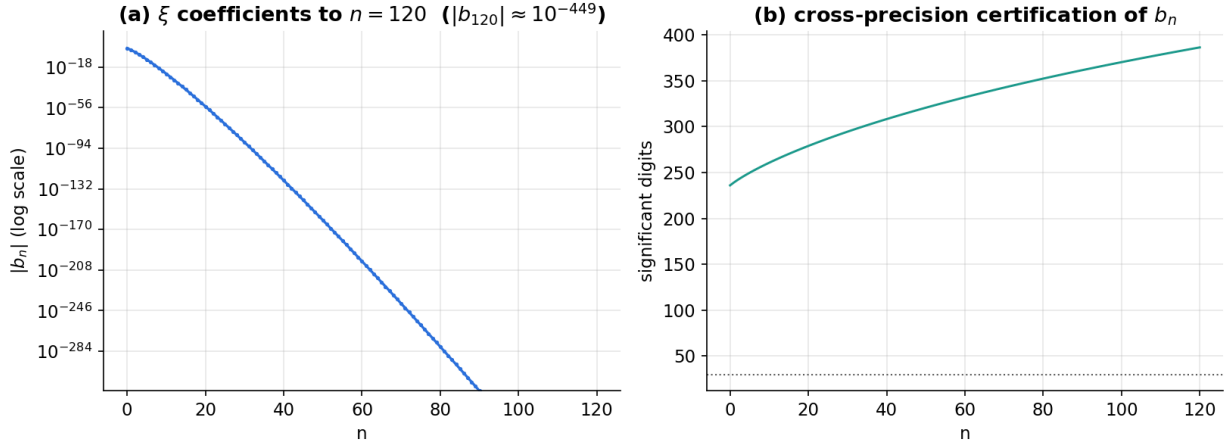


Figure 1: ξ Taylor data to record depth. (a) $|b_n|$ for $n \leq 120$ on a log scale, decaying factorially to $|b_{120}| \approx 10^{-449}$; the alternating sign of b_n gives $\mu_n = (-1)^n b_n > 0$ (order-1 minors, forced by $\Phi > 0$). (b) Cross-precision certification: every b_n agrees to ≥ 236 significant digits between the two independent computations.

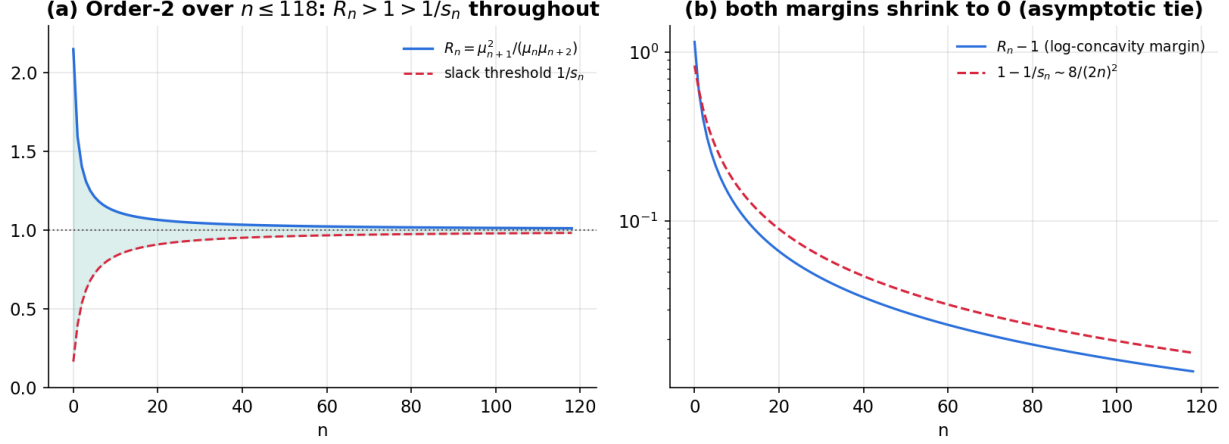


Figure 2: Order-2 over the full range $n \leq 118$. (a) $R_n = \mu_{n+1}^2 / (\mu_n \mu_{n+2}) > 1$ throughout (μ is genuinely log-concave—stronger than the slack requires) and stays above the slack threshold $1/s_n$ (shaded margin); both $R_n \rightarrow 1^+$ and $1/s_n \rightarrow 1^-$. (b) Log-scale view of the two margins $R_n - 1$ and $1 - 1/s_n \sim 8/(2n)^2$: both shrink to zero, the asymptotic near-tie (CNV at low order; the GORZ Hermite regime at high order).

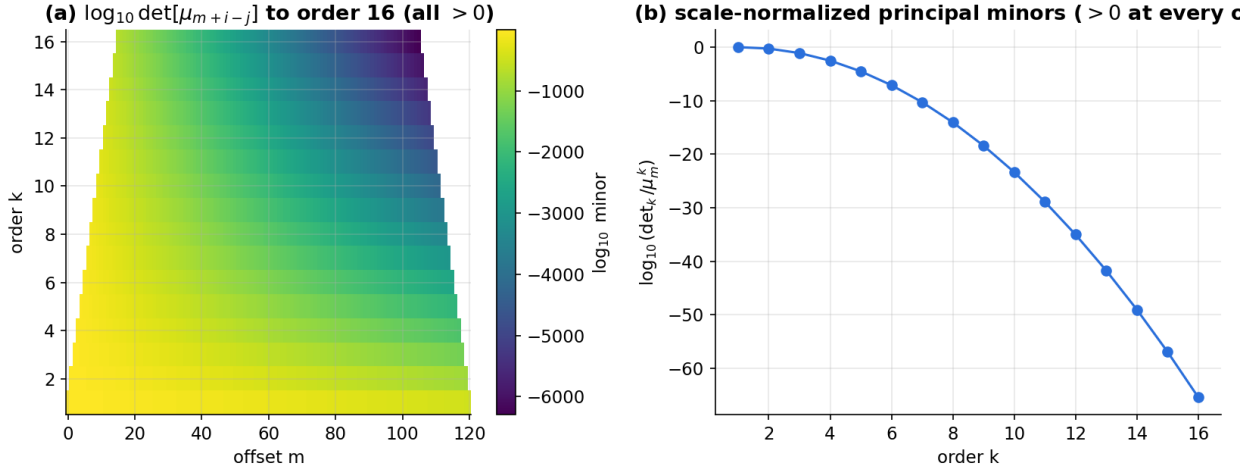


Figure 3: The Pólya-frequency ladder at scale. (a) $\log_{10} \det[\mu_{m+i-j}]$ for orders $k = 1, \dots, 16$ and all offsets m (values span down to $\sim 10^{-6000}$): *every minor is strictly positive*—the smallness is factorial scale, not near-degeneracy. (b) Scale-normalized principal minors $\det_k / \mu_m^k$ remain positive at every order up to 16.

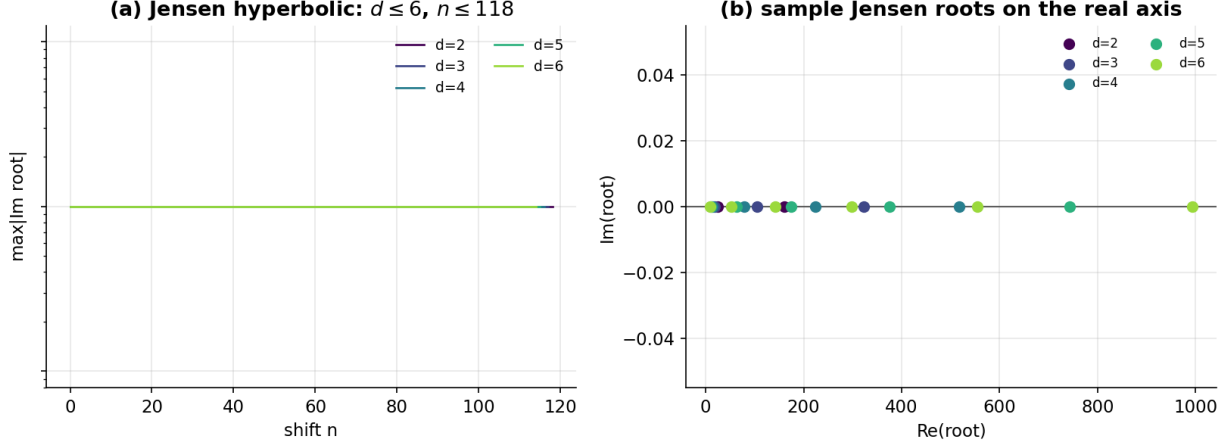


Figure 4: Jensen hyperbolicity at scale. (a) $\max |\operatorname{Im}(\text{root})|$ of $J_{d,n}$ for degrees $d = 2, \dots, 6$ is numerical zero ($\sim 10^{-320}$) across all computed n up to 114. (b) Sample Jensen roots lie on the real axis.

Interpretation. The reduction’s premise holds for ξ at every tested order (Fig. 3), with comfortable low-order margins (Fig. 2); indeed μ is empirically log-concave, strictly stronger than the slack-relaxed Turán requirement. The difficulty is therefore *uniformity*—behaviour as $k \rightarrow \infty$ and $n \rightarrow \infty$ —not any specific low rung.

5 The open frontier

The reductions convert “zeros on a line” into positivity statements about the explicit positive kernel Φ and an explicit operator metric—objects amenable to tools foreign to zero-counting.

- A. Uniform total positivity.** Prove $\det[\mu_{m+i-j}] \geq 0$ for all k, m . Tools: Karlin total positivity [6]; Schoenberg Pólya-frequency *functions*; the variation-diminishing property of the kernel in $\sum \mu_n z^n = \int_0^\infty \Phi(u) \cosh(u\sqrt{z}) du$. (RH-equivalent, by Theorem 5.)
- B. Asymptotic margin.** Control $R_n - 1$ versus $1 - 1/s_n \sim 8/(2n)^2$. Tools: the Griffin–Ono–Rolen–Zagier Hermite asymptotics [8], made effective and uniform in degree.
- C. μ log-concavity from Φ .** Prove $\mu_{n+1}^2 > \mu_n \mu_{n+2}$ (Fig. 2, empirically robust) from the shape of Φ (moment inequalities, log-concavity). Strictly weaker than RH; would close the order-2 rung unconditionally on the corrected coefficients.
- D. Operator existence.** Construct (E, T, η) with a positive metric and spectrum $\{\gamma_k\}$ (operator theory; cf. [11]).

The de Bruijn–Newman programme [9] provides an independent dynamical handle ($\Lambda \geq 0$; RH $\iff \Lambda \leq 0$) on the same object.

6 Methodology, reproducibility, and limitations

Verification. Lean 4 with Mathlib; 47 modules; entry point `Reinmann.lean`. The script `scripts/safe-verify.lean` rejects `sorry`, `admit`, added axioms, and similar shortcuts, and rebuilds all files (exit 0). Every theorem cited above has axiom footprint `{propext, Classical.choice, Quot.sound}`.

Named external inputs (never asserted). `PolyaJensenBridge`, `XiToeplitzTPToScaledFiniteBridge`, `XiLaguerrePolyaClosureBridge`, `XiMomentKernelRep`, `XiToeplitzTPFromRH`. Each is a standard classical theorem; formalizing them in Mathlib is future work.

Limitations.

1. No proof of RH: every RH-implying statement is conditional on an input that is RH-equivalent (Theorem 5) or an unformalized classical theorem.
2. Numerics are finite: the computation certifies positivity/hyperbolicity only for the tested ranges ($n \leq 120$, Toeplitz order $k \leq 16$, Jensen degree $d \leq 6$); however extensive, this is evidence, not proof, and cannot establish the uniform $(k, n \rightarrow \infty)$ statement that is RH-equivalent.
3. The pseudo-Hermitian existence claim is entirely open and unevidenced.
4. Equivalence-chain reformulations (e.g. “ ≤ 1 zero per height”) are correct but RH-equivalent, hence reorganization rather than progress.

7 Related work

Pólya [2, 3]; Aissen–Schoenberg–Whitney [4] and Edrei [5]; Karlin [6]; Csordas–Norfolk–Varga [7]; Griffin–Ono–Rolen–Zagier [8]; de Bruijn–Newman/Rodgers–Tao [9]; Mostafazadeh [10] and Bender–Brody–Müller [11]; Mathlib [12]. Spectral and adelic perspectives (Berry–Keating, Connes) and the Weil/Li positivity criteria are classical antecedents of the operator and positivity viewpoints used here.

Acknowledgement of limitations (summary)

The verified contribution is a correct, tight, axiom-clean *reduction* of RH to an explicit positivity statement, plus an independent operator reduction, a repaired target-function error, and numerical evidence. RH itself remains open, as stated throughout.

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